From classical energy estimates to a conditionally certified ground state: a modular Stetcu–Baroni filter pipeline and its cost for the J_1–J_2 chain

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Classical methods such as the density-matrix renormalization group (DMRG) supply accurate ground-state energies, a gap estimate and a compact trial state, but a shallow circuit for that trial state keeps some weight on excited states. The projection filter of Stetcu, Baroni and Carlson removes it with one ancilla, controlled time evolution and post-selection. We give a generic, modular pipeline that takes energy estimates and an ansatz for any Hamiltonian written as a sum of Pauli strings, designs a cosine filter with a certified suppression ratio, chooses first- or second-order Trotter step counts from commutator bounds, and returns an a-priori bound on the infidelity of the prepared state together with a transparent gate count. The bound is a theorem conditional on the inputs (an interval for the ground energy, a lower bound on the gap and a lower bound on the trial state's ground-space weight), and every one of these conditions is an explicit input. With DMRG inputs these conditions are confirmed by exact diagonalization only for small systems; beyond that the certificate is conditional on the DMRG energies and on an overlap estimated from a reference MPS. We apply it to the open spin-$\frac{1}{2}$ J_1–J_2 chain with DMRG inputs and bond-dimension-χ matrix-product trial states, $N = 8$–48. In all 96 of 96 simulated circuits ($N \leq 14$) the certified bound held; it overestimates the measured infidelity by a median factor of 14, mostly through its Trotter term. A second-order product formula is a median 92× cheaper than first order, and an optimized filter a median 2.9× cheaper than a closed-form halving ("dyadic") schedule, which however is the only design that stays feasible for small overlaps. Near the critical coupling at $N = 48$, a $\chi = 4$ trial state is refined to infidelity 10^{-2} for 9.7×10^6 CNOTs per success. Across all small instances, overestimating the gap by 10–25% or inflating the overlap past $1 - \epsilon$ can break the target, while a conservative gap costs at most 2.9×. For this one-dimensional model at $\epsilon = 10^{-2}$, however, a $\chi = 8$ matrix-product circuit alone already meets the target with about 10^3–10^4 CNOTs, so the value of the pipeline lies in its conditional certificate and in models where compact trial states are not available.

I. INTRODUCTION

A common route to a ground state on a quantum computer starts classically: an approximate method produces a trial state that can be compiled into a short circuit, and a quantum subroutine removes its residual excited-state weight. For one-dimensional and quasi-one-dimensional systems the natural classical engine is DMRG [3, 4], which yields the ground-state energy E_0, excited-state energies and a matrix-product state (MPS) whose low-bond-dimension truncations compile exactly into staircase circuits [5]. The quantum step can be phase estimation [6], a polynomial filter implemented by quantum singular-value transformation [7–9], or a product of cosine filters implemented with a single ancilla [1, 13]. The last option, the projection algorithm of Stetcu, Baroni and Carlson (SBC) [1], needs little beyond time evolution: each step is one controlled time evolution sandwiched between ancilla rotations and a measurement, the filter is designed classically, and the success probability is known in advance. The gate counts we find (10^5–10^9 CNOTs per success) place it on fault-tolerant hardware rather than on noisy devices.

To use such a filter one has to answer three practical questions: which times and phases, how many Trotter steps, and how good the result is. SBC optimize the times and phases numerically for a model spectrum with a given gap; they do not give a bound on the error of the prepared state, and the Trotter error is treated separately [1]. This paper packages the full chain as a pipeline with typed inputs and a certificate:

1. *Inputs*: a Pauli-sum Hamiltonian H, energy estimates (an interval for E_0, a lower bound on the first excited energy E_1, optionally an upper bound on the spectrum) and an ansatz (its estimated ground-space weight γ and the gate cost of its circuit). Any classical method can supply them.

2. *Filter design*: three schedules, from the naive repeated cosine to an optimized design, each with a rigorous interval bound on its suppression ratio.

3. *Trotterization*: first- and second-order commutator-scaling bounds [2], evaluated exactly on the support of each local commutator.

4. *Certificate*: an a-priori bound on the trace distance between the prepared state and the ground space, valid whenever the inputs are valid.

5. *Resources*: qubits, CNOTs, arbitrary rotations, a T-count estimate and the expected cost per successful preparation.

We then apply it to the open J_1–J_2 Heisenberg chain using DMRG, validate the certificate by statevector simulation for $N \leq 14$, and report resources for $N \leq 48$. The code is the `sbpipe` package, available with all data at https://github.com/aaronseymour7/

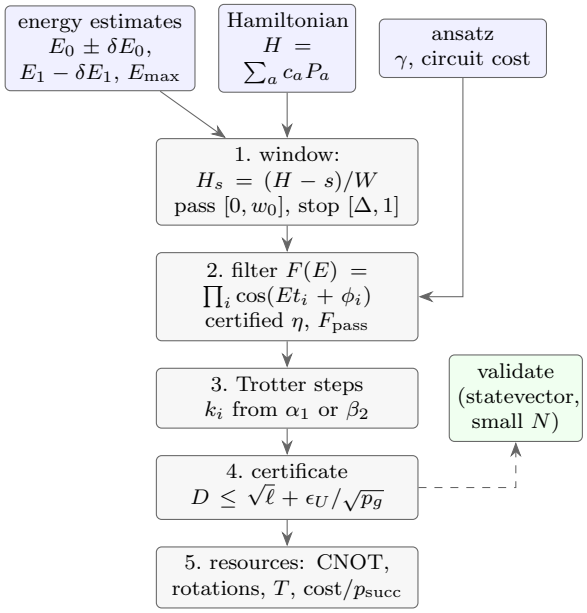


FIG. 1. The pipeline. Blue: inputs, which any classical method can supply. Grey: the stages of `refine()`. Green: the optional statevector check used in Sec. III B.

`FilterPipeline`; every number below is produced by the scripts named in Appendix B.

The contribution is methodological and quantitative rather than algorithmic: the filter is SBC's, the Trotter bounds are those of Childs *et al.* [2], and the certificate follows the standard triangle-inequality argument also used for polynomial filters [7, 9]. End-to-end resource studies that compile a classical (often MPS) trial state and then filter or project it exist for quantum chemistry [11, 12]; here the classical inputs enter only through explicit, typed conditions, and the same accounting is applied to SBC's cosine filters and to QETU. What is new is a generic interface whose guarantees are stated in terms of the inputs, a closed-form "dyadic" energy schedule with $O(\Delta^{-1} \log \eta^{-1})$ total time, and a calibrated account of what this costs and where the bound is loose for a concrete, physically relevant model.

II. THE PIPELINE

A. Inputs and the spectral window

The Hamiltonian is $H = \sum_a c_a P_a$ with real c_a and Pauli strings P_a, in a fixed order that is also the Trotter order, and with an optional grouping of terms (here: one group per bond). The pipeline assumes

$$E_0 \in [E_0^{\text{est}} - \delta E_0,\ E_0^{\text{est}} + \delta E_0], \qquad E_1 \ge E_1^{\text{est}} - \delta E_1, \qquad E_{\text{top}} \le E_{\max}, \tag{1}$$

where E_1 is the lowest level above the ground space that the trial state can overlap with. When the trial state's quantum numbers are unknown, as for the DMRG trial states below, which are not symmetry-adapted, this is the global first excited level; restricting to the trial state's symmetry sector could only raise it. The certificate is a statement about the actual Hamiltonian only if (1) holds for it: with an overestimated E_1 a real excitation can sit in the filter's transition region, and the bound then says nothing (Sec. III D). If no $E_{\max}$ is given, the Weyl bound $E_{\max} = \sum_g \lambda_{\max}(h_g)$ over the term groups h_g is used; it is rigorous and, for the antiferromagnetic J_1–J_2 chain, attained by the ferromagnetic state. With $s = E_0^{\text{est}} - \delta E_0$ and $W = E_{\max} - s$, the scaled Hamiltonian $H_s = (H - s)/W$ has its ground energy in the *pass band* $[0, w_0]$, $w_0 = 2\delta E_0/W$, and every other relevant level in the *stop band* $[\Delta, 1]$, $\Delta = (E_1^{\text{est}} - \delta E_1 - s)/W$.

The ansatz enters through γ, a lower estimate of its weight on the ground space, and through the gate cost of the circuit that prepares it. A statevector is needed only for validation.

B. The filter and three schedules

One SBC pulse [1] applies $\mathsf{H}_a\, e^{-itH_s \otimes Z_a}\, R_z^{(a)}(2\phi)\, \mathsf{H}_a$ to the system and an ancilla in $|0\rangle$; the ancilla-$|0\rangle$ block of this unitary is $\cos(H_s t + \phi)$. (SBC write the same block with $e^{-i(tH+\delta)\otimes Y_a}$.) Keeping outcome 0 after m pulses applies

$$F(H_s) = \prod_{i=1}^{m} \cos(H_s t_i + \phi_i), \qquad T = \sum_i |t_i|. \tag{2}$$

Two numbers characterize a filter: the pass-band floor $F_{\text{pass}} \le \min_{[0,w_0]} |F|$ and the suppression ratio $\eta = \sup_{[\Delta,1]} |F|/F_{\text{pass}}$. Both are certified, not sampled: since $|F'| \le L := \sum_i |t_i|$ and $|F''| \le L^2$, an interval branch-and-bound gives a rigorous upper bound on $\sup |F|$ over the stop band, and $F_{\text{pass}} = |F(0)| - Lw_0$ (the certifier is `sbpipe.floor.certify`; it is used for the optimized schedule, the other two have closed-form certificates). Given a target leakage ϵ_ℓ and γ, the design target is $\eta^2 \le \gamma\epsilon_\ell/[(1-\epsilon_\ell)(1-\gamma)]$ (Lemma 1). We provide three schedules:

Uniform: m equal pulses $t = \pi/(1+\Delta)$, $\phi = 0$, so $F = \cos^m(Et)$. For $E \in [\Delta, 1]$ we have $Et \in [\theta, \pi - \theta]$, $\theta = \pi\Delta/(1+\Delta)$, hence $\sup_{[\Delta,1]} |\cos Et| = \cos\theta$. Then $m \approx 2\ln(1/\eta)/\theta^2$ and $T \approx 2\ln(1/\eta)/(\pi\Delta^2)$: quadratic in the inverse gap. It is the baseline that any design should beat.

Dyadic: r repetitions of the halving ladder $t = \tau, 2\tau, \ldots, 2^{K-1}\tau$, $\tau = \pi/(1+\Delta)$, $\phi = 0$. This is the schedule SBC use for quantum-number projection [1], here applied to the energy. One block is a Dirichlet kernel,

$$\prod_{k=0}^{K-1} \cos(2^k E\tau) = \frac{\sin(2^K E\tau)}{2^K \sin(E\tau)}, \tag{3}$$

which equals 1 at $E = 0$. The identity is exact for every E (repeated use of $\sin 2x = 2\sin x \cos x$). The suppression bound is a separate step that uses the spectral window: for $E \in [\Delta, 1]$, $E\tau \in [\theta, \pi - \theta]$, so $\sin(E\tau) \geq \sin\theta$ and, with $|\sin(2^K E\tau)| \leq 1$, the block is at most $\eta_b = 1/(2^K \sin\theta)$ in absolute value there. This bound relies on the scaling H_s, which places every relevant level in $[0, 1]$, and on the choice of τ; it does not hold outside $[\Delta, 1]$. With $r = \lceil \ln\eta / \ln\eta_b \rceil$ blocks, $T = r\tau(2^K - 1)$; choosing $2^K \sin\theta \approx e$ gives

$$T_{\text{dyadic}} \approx \frac{e\,\tau}{\sin\theta} \ln\frac{1}{\eta} \;\approx\; \frac{e}{\Delta} \ln\frac{1}{\eta}, \tag{4}$$

the optimal $O(\Delta^{-1} \log \eta^{-1})$ scaling [7] with an explicit constant and no optimization. The pipeline searches K and r for the smallest T. Both closed-form schedules are certified by the inequalities just given (with $F_{\text{pass}} = \cos^m(w_0 t)$ and $1 - Lw_0$, respectively), which avoids interval arithmetic on filters with $L \sim 10^3$–10^5; the test suite checks them against a 2×10^6-point grid.

Optimized: times and phases chosen to maximize $F(0)^2$ subject to the certified constraint on η at fixed (T, m), over a grid of $x = T\Delta/\pi$ and $m \in \{4, 6, 8\}$, picking the pair with the smallest $T^2/F(0)^2$. This is SBC's recipe for energy filters [1] with a certified constraint in place of a grid objective; it uses the SLSQP designer `sbpipe/floor.py`. Nonzero phases trade success probability ($F(0) < 1$) for a shorter T.

QETU baseline. To compare with a polynomial filter under the same accounting, the pipeline also designs the quantum eigenvalue transformation of unitaries (QETU) of Dong, Lin and Tong [9]. It uses the same primitive, d queries of $e^{-i\tau H_s \otimes Z_a}$, the controlled-reversal form also used by the rodeo algorithm [10], but interleaves them with ancilla rotations and post-selects only once. On every eigenvector the ancilla then undergoes quantum signal processing in $x = \cos(\tau E)$, so the ancilla-0 block is $p(\cos\tau H_s)$ for any real even polynomial p of degree d with $|p| \leq 1$ on $[-1, 1]$ [9]. With $\tau = \pi/2$ the scaled spectrum maps to $x \in [0, 1]$ and the stop band to $[0, x_s]$, $x_s = \cos(\pi\Delta/2)$. We take the Chebyshev filter

$$p(x) = \frac{T_d(x/x_s)}{T_d(1/x_s)}, \qquad \eta = \frac{1}{T_d(\cos(\tau w_0)/x_s)}, \tag{5}$$

which is extremal: no polynomial of degree d bounded by 1 is smaller on the stop band relative to its value at the pass band. It is certified in closed form (p is monotone on $[x_s, 1]$ and equioscillates on $[-x_s, x_s]$), has $p(0) = 1$ so $p_g = \gamma F_{\text{pass}}^2 \approx \gamma$, and needs $T = d\tau \approx \Delta^{-1} \ln(2/\eta)$, two to three times less than the dyadic ladder. The Trotter bound, step allocation and Theorem 1 carry over unchanged, with one pulse per query and two ancilla rotations per query (the phase and the identity term). The QSP phases are computed classically; our cost model does not include that step. For second-order formulas $S_2(-\tau) = S_2(\tau)^\dagger$, so the Trotterized circuit implements exactly $p(\tilde{X})$ with $\tilde{X} = \frac{1}{2}(S_2(\tau) + S_2(-\tau))$, which is how we simulate it; the test suite checks this against an explicit ancilla circuit with QSP phases.

For each schedule the leakage share $\epsilon_\ell/\epsilon \in \{0.5, 0.25, 0.1, 0.05\}$ is scanned and the plan with the smallest expected CNOT count per success is kept. An "auto" mode does the same over the dyadic and optimized schedules together.

C. Trotterization

Each controlled evolution $e^{-it_i H_s \otimes Z_a}$ is approximated by k_i steps of the first-order (Lie) or second-order (Strang) product formula over the non-identity terms; the identity term of H_s becomes a single ancilla R_z per pulse and is exact. For $B_a = \sum_{b>a} H_b$ the bounds of Childs *et al.* [2, Props. 15–16] are

$$\|S_1(t) - e^{-itH}\| \leq \tfrac{t^2}{2}\alpha_1, \quad \alpha_1 = \textstyle\sum_a \|[B_a, H_a]\|, \tag{6}$$

$$\|S_2(t) - e^{-itH}\| \leq t^3\beta_2, \quad \beta_2 = \tfrac{1}{12}\textstyle\sum_a \|[B_a,[B_a,H_a]]\| + \tfrac{1}{24}\textstyle\sum_a \|[H_a,[H_a,B_a]]\|, \tag{7}$$

so k steps over time t cost $\alpha_1 t^2/(2k)$ or $\beta_2 t^3/k^2$. The filter times t_i are in units of the scaled Hamiltonian, so the constants are evaluated for the terms of H_s, i.e. for coefficients c_a/W (the identity shift does not enter any commutator). Equivalently $\alpha_1(H_s) = \alpha_1(H)/W^2$ and $\beta_2(H_s) = \beta_2(H)/W^3$, with physical evolution time t_i/W; the code uses the first convention, and we checked that it agrees with the second to machine precision. Tensoring every term with Z_a leaves α_1 and β_2 unchanged because $Z_a^2 = 1$. For Pauli strings $[P_b, P_a] = 2P_bP_a$ when they anticommute and 0 otherwise, so each commutator is a short Pauli sum supported on a few qubits around term a; we evaluate its norm exactly on that support (dense for ≤ 8 qubits, Lanczos to a relative tolerance of 10^{-12} up to 12 with a 10^{-9} safety factor, triangle inequality beyond, which never occurred here). The Lanczos values are numerically exact but not certified upper bounds; replacing them by the triangle inequality would make every constant rigorous at a higher cost. The constants are maximized over the forward and reversed term order, so they hold for either convention. Every pulse evolves under the same $H_s \otimes Z_a$, so the per-pulse bounds $\alpha_1 t_i^2/(2k_i)$ and $\beta_2|t_i|^3/k_i^2$ share one constant; the phase ϕ_i enters only the ancilla rotation, and the bound is symmetric in the sign of t_i. With equal constants, the continuous relaxation of $\min \sum_i k_i$ subject to $\sum_i(\text{bound}_i) \leq \epsilon_T$ is solved by $k_i \propto |t_i|$ for both orders (Lagrange). We take $k_i = \lceil |t_i|/\delta t \rceil$ with the common step δt of that solution. Rounding up keeps the bound, costs at most m steps more than the continuous optimum and need not be the integer optimum; the budget it frees goes to rotation synthesis. The certified times are untouched.

D. The certificate

Theorem 1. *Assume (1) and that the trial state has ground-space weight at least γ. Let the designed filter have certified η and F_{pass}, let ϵ_U bound the operator-norm error of the implementation (Trotter error $\sum_i \epsilon_i$ plus $R\delta$ for the R arbitrary rotations of the trial and filter circuits, each synthesized to accuracy δ), and let*

$$\ell = \frac{(1-\gamma)\eta^2}{\gamma + (1-\gamma)\eta^2}, \qquad p_g = \gamma F_{\mathrm{pass}}^2. \tag{8}$$

Let $|g\rangle = \Pi_0|\psi\rangle / \|\Pi_0|\psi\rangle\|$ be the normalized projection of the trial state onto the ground space, with Π_0 the ground-space projector; $|g\rangle$ is a pure state also when the ground space is degenerate. For a noiseless implementation the state $|\tilde{\psi}_{\mathrm{out}}\rangle$ prepared conditioned on the ancilla outcomes 0 is pure, and

$$D \le \sqrt{\ell} + \frac{\epsilon_U}{\sqrt{p_g}}, \qquad 1 - \mathrm{Tr}(\Pi_0 \tilde{\rho}) \le 1 - |\langle g|\tilde{\psi}_{\mathrm{out}}\rangle|^2 = D^2, \tag{9}$$

where D is the trace distance between $\tilde{\rho} = |\tilde{\psi}_{\mathrm{out}}\rangle\langle\tilde{\psi}_{\mathrm{out}}|$ and $|g\rangle$. Moreover $\langle H\rangle - E_0 \le (E_{\mathrm{top}} - E_0)D^2$, and the success probability is at least $(\sqrt{p_g} - \epsilon_U)^2$. Below, $1 - F$ denotes $1 - |\langle g|\tilde{\psi}_{\mathrm{out}}\rangle|^2$; for the nondegenerate ground state of the chains studied here it is the usual infidelity.

The proof (Appendix A) combines a leakage lemma for the exact filter with the observation that a vector error ϵ_U on an unnormalized vector of norm at least $\sqrt{p_g}$ moves its direction by at most $\epsilon_U/\sqrt{p_g}$. The pipeline sets the budget $\epsilon_U \le (\sqrt{\epsilon} - \sqrt{\ell})\sqrt{p_g}$, gives 90% of it to the product formula and the rest to rotation synthesis. If the trial state already satisfies $1 - \gamma \le \epsilon$, the pipeline also offers the empty filter ($\eta = 1$, no post-selection, $\ell = 1-\gamma$) and keeps whichever plan is cheaper.

E. Resource model

All-to-all connectivity and the Clifford+R_z gate set are assumed. A term $cP \otimes Z_a$ with $|P| = w$ is a weight-$(w+1)$ Pauli rotation: $2w$ CNOTs in a ladder and one R_z. A Lie step costs one rotation per term; k Strang steps cost $k(2\Gamma - 2) + 1$ rotations for Γ terms because the two middle half-steps and the outer half-steps of adjacent steps merge. Each pulse adds one ancilla R_z. An arbitrary rotation at accuracy δ costs $3\log_2(1/\delta)$ T gates, the leading term of Ross–Selinger synthesis [14]. The cost of one attempt is the trial circuit plus the filter, and the expected cost per success is the attempt cost divided by the certified lower bound $(\sqrt{p_g} - \epsilon_U)^2$ on the success probability of the implemented circuit from Theorem 1 (repeat-until-success; an early-abort protocol that measures after each pulse would be cheaper and is not modeled). The pipeline spends the whole budget $\epsilon_U = (\sqrt{\epsilon} - \sqrt{\ell})\sqrt{p_g}$, so this denominator is $p_g(1 - \sqrt{\epsilon} + \sqrt{\ell})^2 \ge (1 - \sqrt{\epsilon})^2 p_g$, i.e. at least $0.81\, p_g$ at $\epsilon = 10^{-2}$. Without a filter nothing is post-selected and every attempt succeeds. Alongside each T count we report the number of arbitrary rotations per attempt and their common synthesis accuracy δ. The ladder CNOT count is an upper bound: qiskit's synthesis shares CNOTs between the XX, YY, ZZ terms of a bond and needs 8 instead of 12 per controlled bond per Lie step (checked in the test suite), so CNOT numbers below can be read as upper bounds that a compiler lowers by about one third.

III. APPLICATION: J_1–J_2 CHAIN WITH DMRG

A. Model and front end

We study the open spin-$\frac{1}{2}$ chain

$$H = J_1 \sum_{i=1}^{N-1} \mathbf{S}_i \cdot \mathbf{S}_{i+1} + J_2 \sum_{i=1}^{N-2} \mathbf{S}_i \cdot \mathbf{S}_{i+2}, \tag{10}$$

$J_1 = 1$, at $J_2 = 0$ (Heisenberg), $J_2 = 0.2411$ (close to the critical point of the gapless-to-dimerized transition [16, 17]) and $J_2 = 0.4$ (dimerized), for $N = 8$ to 48. Its Pauli form has $3(N-1)$ terms of weight 2 at $J_2 = 0$ and $3(2N-3)$ otherwise. Terms are ordered by bond: nearest-neighbor bonds with even then odd left site, then next-nearest-neighbor bonds likewise.

The front end (`sbpipe/frontends/dmrg_quimb.py`) builds the MPO from the Pauli sum and runs two-site DMRG (quimb [18]) with bond dimensions up to 256 ($N \le 24$) or 128 ($N > 24$) and truncation cutoff 10^{-12}. E_1 is the lowest state of $H + \mu(S^z_{\mathrm{tot}} - 1)^2$, $\mu = 2$, i.e. the lowest triplet, which we use as the global first excitation. The error bars $\delta E_0, \delta E_1$ are the energy change between the last two bond dimensions, floored at 10^{-10}; they are convergence heuristics, not bounds. Both DMRG energies are variational upper bounds, so a residual DMRG error in E_1 (or convergence of the penalty run to a higher state) biases the gap upward, which is the dangerous direction (Sec. III D); these error bars do not protect against that bias, and the certificate for $N > 14$ is conditional on the gap. The trial state is the variationally optimal MPS at fixed bond dimension $\chi \in \{1, 2, 4, 8\}$, and γ is its overlap with the reference MPS. This is an estimate of the overlap with the exact ground state, not a lower bound on it, so for $N > 14$ the certificate is equally conditional on γ. An MPS of bond dimension 2^b is exactly a staircase of $N - b$ unitaries on $b+1$ qubits; we cost each with the quantum Shannon decomposition bound $\frac{23}{48}4^n - \frac{3}{2}2^n + \frac{4}{3}$ CNOTs (3, 20, 100 for $n = 2, 3, 4$) [15] and $4^n - 1$ rotations.

For $N \le 14$ we compare with exact diagonalization (ED). DMRG reproduces E_0 to 2.3×10^{-11}, the first excited level of any symmetry to 2.4×10^{-11} (so the first excitation is the triplet in every ED case), and the trial overlaps to 4.0×10^{-8} (largest difference, for a $\chi = 1$

TABLE I. DMRG inputs: ground energy, gap $E_1 - E_0$ and trial overlap γ of the optimal bond-dimension-χ MPS. Source: `results/inputs.json`.

N	J_2	E_0	gap	$\gamma_{\chi=1}$	γ_2	γ_4	γ_8
8	0	-3.374933	0.3927	0.134	0.775	0.999	1.0000
12	0	-5.142091	0.2809	0.060	0.591	0.994	1.0000
16	0	-6.911737	0.2193	0.028	0.475	0.984	0.9999
24	0	-10.453786	0.1530	0.006	0.328	0.954	0.9994
32	0	-13.997316	0.1177	0.001	0.234	0.913	0.9978
48	0	-21.085956	0.0809	0.000	0.125	0.817	0.9902
8	0.2411	-3.135001	0.3859	0.113	0.904	1.000	1.0000
12	0.2411	-4.738077	0.2720	0.046	0.806	1.000	1.0000
16	0.2411	-6.343316	0.2100	0.019	0.703	0.998	1.0000
24	0.2411	-9.556221	0.1443	0.003	0.517	0.989	0.9997
32	0.2411	-12.770432	0.1099	0.001	0.373	0.972	0.9989
48	0.2411	-19.200243	0.0744	0.000	0.191	0.920	0.9946
8	0.4	-3.025191	0.4037	0.036	0.976	1.000	1.0000
12	0.4	-4.545626	0.2956	0.003	0.945	1.000	1.0000
16	0.4	-6.066666	0.2371	0.000	0.911	0.999	1.0000
24	0.4	-9.109325	0.1748	0.000	0.839	0.995	0.9998
32	0.4	-12.152237	0.1418	0.000	0.768	0.990	0.9995
48	0.4	-18.238247	0.1075	0.000	0.640	0.976	0.9981

trial). The Weyl bound equals the exact top of the spectrum to 1.2×10^{-14}. Table I lists the inputs. The trial overlap decays roughly exponentially with N at fixed χ, because a fixed bond dimension captures a fixed amount of entanglement per bond while the truncation error accumulates over N bonds, so the filter has more to do at larger N.

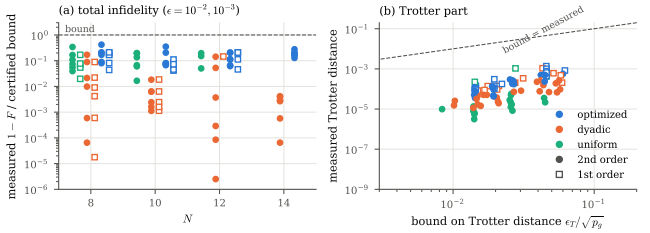


FIG. 2. All simulated plans. (a) Measured infidelity of the planned circuit divided by its certified bound; points below the dashed line respect the bound and are offset horizontally by schedule and order. (b) Measured distance between the Trotterized and the exact filtered state against its certified bound. Points below the dashed lines respect the bound. Filled circles: second order; open squares: first order.

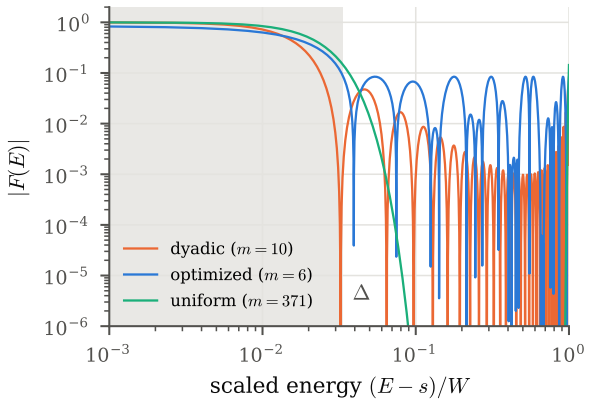


FIG. 3. $|F(E)|$ of the three schedules for $N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$, second order. Dotted lines: certified ηF_{pass}. Grey: the gap region. The horizontal axis is linear below Δ and logarithmic above.

B. Validation of the certificate ($N \leq 14$)

For $N \in \{8, 10, 12, 14\}$, the three couplings, $\epsilon \in \{10^{-2}, 10^{-3}\}$, the three schedules and both Trotter orders we plan a circuit for the $\chi = 2$ trial state and, when its size permits, simulate it exactly: the ancilla-0 block of the Trotterized circuit is $\frac{1}{2}(e^{-i\phi}S(t) + e^{i\phi}S(-t))$ per pulse, applied to the trial statevector and compared with the ED ground state. Of 144 plans, 96 were simulated (the others, mostly uniform or first-order plans with 10^4–10^8 steps, exceed our simulation budget), and the certificate held in all 96 (Fig. 2).

The bound is conservative by a median factor 14 (range 2.4–4.0 $\times$ 10^5) in infidelity. Splitting it shows where: the leakage term ℓ overestimates the exact filter's infidelity by a median 4.3$\times$ (range 1.1–1.3 $\times 10^5$). By schedule the medians are 1.4 (optimized), 4 (uniform) and 54 (dyadic): the optimized design pushes $|F|$ to ηF_{pass} across the band, whereas the closed-form dyadic bound $1/(2^K \sin\theta)$ is reached only near the band edge. More generally, η is a worst case over the whole stop band while the trial state's excited weight sits mostly low in the band. The Trotter term overestimates the measured Trotter distance by a median 185$\times$ (range 26–4.5 $\times 10^3$): commutator bounds are worst-case over all states, while the filtered state lives in the low-energy subspace, where Trotter errors are much smaller [19]. The measured success probability equals p_g to two digits in every case, because after filtering almost all of the surviving norm is ground-state weight.

Table II compares the schedules for one case. The uniform filter needs a total time an order of magnitude larger than the others, as its Δ^{-2} scaling predicts. The optimized filter is shortest; its nonzero phases lower p_g (0.58 against 0.81 here), which the cost criterion accounts for. Second order cuts the step count by one to two orders of magnitude at these sizes. Figure 3 shows the three filter functions.

How much of the remaining gap could a better bound recover? For $N \leq 12$ we bisect a common scale factor on

TABLE II. Plans for $N = 12$, $J_2 = 0.2411$, $\chi = 2$ trial. $T_{\rm phys} = T/W$ in units of $1/J_1$; CNOT and T counts per attempt (filter) and expected per success (trial + filter, divided by the certified success probability $(\sqrt{p_g} - \epsilon_U)^2$). "rot.": arbitrary rotations per attempt (trial + filter), each synthesized to accuracy δ. "meas.": simulated infidelity of the planned circuit. Source: `results/validate.json`.

ϵ	schedule	order	m	$T_{\rm phys}$	p_g	steps	CNOT (filter)	CNOT/success	rot.	δ	T/success	meas. $1-F$
10^{-2}	uniform	1	437	164.2	0.81	4,136,205	1.0×10^{9}	1.4×10^{9}	2.6×10^{8}	2×10^{-11}	3.8×10^{10}	–
10^{-2}	uniform	2	371	139.4	0.81	17,437	8.7×10^{6}	1.1×10^{7}	2.2×10^{6}	1×10^{-9}	2.5×10^{8}	1.5×10^{-3}
10^{-2}	dyadic	1	10	23.3	0.81	77,308	1.9×10^{7}	2.7×10^{7}	4.9×10^{6}	1×10^{-9}	6.1×10^{8}	–
10^{-2}	dyadic	2	10	23.3	0.81	884	4.4×10^{5}	6.1×10^{5}	1.1×10^{5}	5×10^{-8}	1.1×10^{7}	8.5×10^{-7}
10^{-2}	optimized	1	6	8.7	0.58	13,384	3.4×10^{6}	6.5×10^{6}	8.4×10^{5}	5×10^{-9}	1.3×10^{8}	1.1×10^{-3}
10^{-2}	optimized	2	6	8.7	0.58	226	1.1×10^{5}	2.2×10^{5}	2.8×10^{4}	2×10^{-7}	3.7×10^{6}	1.1×10^{-3}
10^{-3}	uniform	1	745	279.9	0.81	27,817,555	7.0×10^{9}	9.1×10^{9}	1.8×10^{9}	1×10^{-12}	2.7×10^{11}	–
10^{-3}	uniform	2	658	247.2	0.81	55,930	2.8×10^{7}	3.6×10^{7}	6.9×10^{6}	2×10^{-10}	8.6×10^{8}	–
10^{-3}	dyadic	1	15	34.9	0.81	536,160	1.4×10^{8}	1.7×10^{8}	3.4×10^{7}	5×10^{-11}	4.5×10^{9}	–
10^{-3}	dyadic	2	15	34.9	0.81	2,835	1.4×10^{6}	1.8×10^{6}	3.5×10^{5}	5×10^{-9}	3.8×10^{7}	2.5×10^{-9}
10^{-3}	optimized	1	8	14.4	0.67	80,421	2.0×10^{7}	3.1×10^{7}	5.1×10^{6}	4×10^{-10}	7.4×10^{8}	–
10^{-3}	optimized	2	8	14.4	0.67	709	3.5×10^{5}	5.5×10^{5}	8.8×10^{4}	2×10^{-8}	1.0×10^{7}	6.2×10^{-5}

TABLE III. Certified over oracle step count (second order, $\chi = 2$). Columns: $N = 8, 10, 12$ for each J_2. Source: `results/oracle.json`.

		$J_2 = 0$			$J_2 = 0.2411$			$J_2 = 0.4$		
ϵ	schedule	8	10	12	8	10	12	8	10	12
1×10^{-2}	dyadic	13	22	30	24	24	34	18	15	28
1×10^{-2}	optimized	9	12	13	14	16	21	11	12	17
1×10^{-3}	dyadic	22	24	39	31	36	39	20	27	28
1×10^{-3}	optimized	12	17	21	24	22	27	14	16	17

the certified k_i until the simulated infidelity just meets ϵ. The certified step count is 8.7–39× (median 21) larger than this oracle value (Table III). The oracle uses the exact ground state and is not a usable rule; it measures the price of certification at second order.

C. Resources up to $N = 48$

Without simulation the pipeline only needs DMRG, so we plan for N up to 48 with the same $\chi = 2$ trial family (Fig. 4, Table IV). Over all sizes, couplings and targets the second-order plan is 16–661× (median 92) cheaper than the first-order plan of the same schedule. The optimized schedule is 0.7–3.7× (median 2.9) cheaper than the dyadic one at second order, and more expensive in 1 of 29 comparable cases. Its numerical design, however, failed in 14 of 144 plans, all at $N \geq 32$ where $\gamma \lesssim 0.4$ or $\epsilon = 10^{-3}$ demand a suppression η the SLSQP search with $F(0)^2 \geq 10^{-3}$ did not reach; the dyadic schedule is feasible everywhere, which is why the pipeline offers an automatic choice between the two. For $N = 16$–48 the expected CNOT count of the dyadic second-order plan grows as $N^{4.9}$, $N^{5.0}$ and $N^{3.5}$ for $J_2 = 0, 0.2411, 0.4$, and its step count as $N^{2.6}$, $N^{2.7}$, $N^{2.1}$. These exponents combine the shrinking gap ($T \propto 1/\Delta$, and $\Delta = g/W$ with $W \propto N$), the growing commutator constant, the number of terms per step and the decaying overlap (through the success probability); they are fits over $N = 16$–48, not asymptotic statements. At $N = 48$, $J_2 = 0.2411$ the best $\chi = 2$ plan (dyadic) needs 5.4×10^{8} CNOTs per success, which the next section reduces by choosing a better trial.

Comparison with QETU. The Chebyshev QETU filter, planned with the same Trotter, synthesis and cost model, is feasible in all 36 second-order cases (0 failures) and is a median 3.3× (range 2.1–3.9) cheaper than the dyadic schedule. Against the optimized SBC design, where that design exists, the ratio is a median 1.00 (range 0.85–4.3, 29 cases): the two cost the same to within the spread, because both reach $T \approx \Delta^{-1} \ln(1/\eta)$ up to a constant and the Trotter cost depends only on T. At $N = 48$, $J_2 = 0.2411$ QETU needs 1.6×10^{8} CNOTs per success against 5.4×10^{8} for the dyadic plan (Fig. 4). Simulated as in Sec. III B, the certificate held in 24 of 24 QETU plans ($N \leq 14$, both targets), with a median looseness of 7.5×; its leakage term is tight (median 1.3×) because the Chebyshev filter equioscillates across the stop band. The SBC filter therefore offers no cost advantage over a polynomial filter; what it offers is the absence of a phase-finding step and a measurement after every pulse, which an early-abort protocol (not modeled here) could exploit. The pipeline treats both as interchangeable back ends.

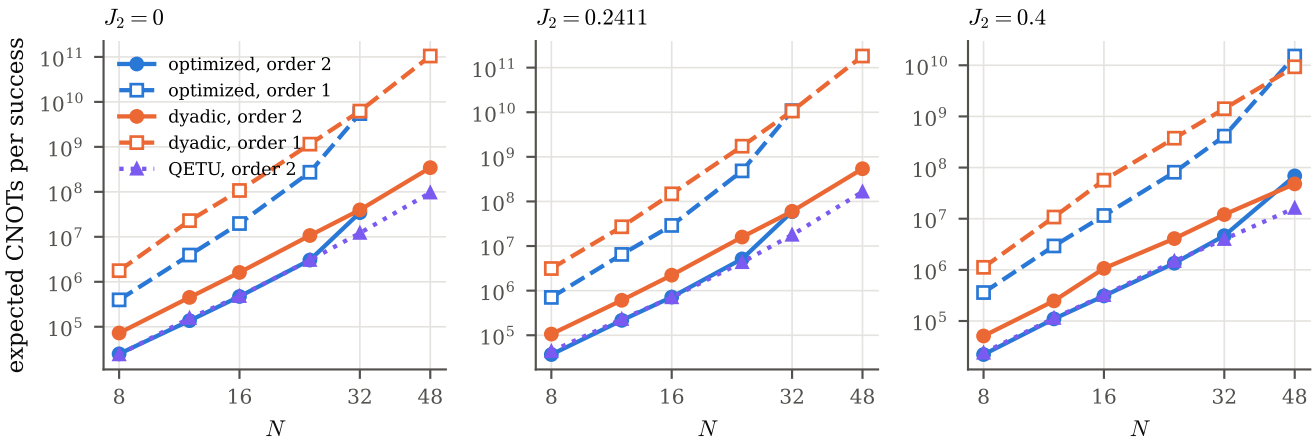


FIG. 4. Expected CNOTs per successful preparation versus N (trial $\chi = 2$, $\epsilon = 10^{-2}$), including the trial circuit and the repetitions (certified success probability). Solid: second order; dashed: first order. Missing optimized points: no feasible design (see text). Dotted triangles: QETU, second order. Sources: `results/scaling.json`, `results/ext_qetu_scaling.json`.

TABLE IV. Cheapest second-order plan ("sch.": dyadic or optimized) for $\epsilon = 10^{-2}$, trial $\chi = 2$. Δ is the scaled gap, $T_{\text{phys}} = T/W$ in $1/J_1$. Steps for first (1st) and second (2nd) order with the same schedule; CNOT and T counts for second order per attempt (att.) and per success (succ.), divided by the certified success probability; "rot.": arbitrary rotations per attempt, synthesized to accuracy δ. Source: `results/scaling.json`.

N	J_2	γ	Δ	sch.	T_{phys}	p_g	steps (1st)	steps (2nd)	CNOT/att.	CNOT/succ.	rot.	δ	T/succ.
16	0	0.475	0.0206	opt.	14.3	0.38	38,020	464	1.6×10^5	4.7×10^5	4.1×10^4	9×10^{-8}	8.3×10^6
16	0.2411	0.703	0.0192	opt.	15.0	0.58	41,572	521	3.6×10^5	7.2×10^5	9.0×10^4	7×10^{-8}	1.3×10^7
16	0.4	0.911	0.0211	opt.	6.6	0.63	19,266	264	1.8×10^5	3.1×10^5	4.6×10^4	6×10^{-8}	5.6×10^6
24	0	0.328	0.0094	opt.	20.5	0.27	251,160	1,422	7.7×10^5	3.0×10^6	1.9×10^5	8×10^{-9}	6.1×10^7
24	0.2411	0.517	0.0087	opt.	16.3	0.29	248,256	1,329	1.4×10^6	5.1×10^6	3.6×10^5	5×10^{-9}	1.1×10^8
24	0.4	0.839	0.0102	opt.	10.8	0.59	87,533	700	7.5×10^5	1.3×10^6	1.9×10^5	1×10^{-8}	2.6×10^7
32	0	0.234	0.0054	opt.	80.0	0.23	3,103,919	9,847	7.2×10^6	3.4×10^7	1.8×10^6	1×10^{-9}	7.6×10^8
32	0.2411	0.373	0.0049	opt.	85.8	0.37	4,979,405	13,617	2.0×10^7	5.9×10^7	5.0×10^6	6×10^{-10}	1.4×10^9
32	0.4	0.768	0.0062	opt.	13.3	0.41	221,517	1,228	1.8×10^6	4.7×10^6	4.5×10^5	5×10^{-9}	9.7×10^7
48	0	0.125	0.0025	dya.	146.0	0.13	20,734,266	34,380	3.9×10^7	3.4×10^8	9.6×10^6	2×10^{-10}	8.3×10^9
48	0.2411	0.191	0.0022	dya.	142.5	0.19	27,888,279	41,586	9.2×10^7	5.4×10^8	2.3×10^7	1×10^{-10}	1.3×10^{10}
48	0.4	0.640	0.0031	dya.	69.3	0.64	4,816,485	12,414	2.8×10^7	4.8×10^7	6.9×10^6	6×10^{-10}	1.1×10^9

D. What if the energy estimates are wrong?

The certificate is only as good as its inputs, so we perturb them for $N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$ (optimized, second order) and simulate the resulting circuit on the true trial state (Fig. 5).

Gap. A gap assumed larger than the true one moves the stop-band edge above the first excitation, which then sits in the filter's transition region. The target was missed in 3 cases, $f \geq 1.5$, with infidelity up to 5.1×10^{-2} at $f = 3$; at $f = 1.1$ and 1.25 the looseness of the certificate still covered the error. Underestimating the gap is safe and cheap: even $f = 0.5$ costs only 2.9×. Since DMRG gives an *upper* bound on E_1, an explicit margin δE_1 is the natural safeguard.

Ground energy. An error bar δE_0 widens the pass band and lowers F_{pass}; the cost stays flat until δE_0 is a sizeable fraction of the gap (13.5× at $\delta E_0 = 0.1$ gap). DMRG energies are far more accurate than this, so E_0 is never the limiting input here.

Overlap. Underestimating γ is safe (a stricter η); the planned cost rises 9.2× at $f = 0.25$, mostly because the planned success probability $p_g = \gamma F_{\text{pass}}^2$ uses the pessimistic γ, while the actual success rate does not change. Overestimating it by 2% did not break the target in this instance (0 failures), because the leakage term is itself loose.

All instances. To see whether this one instance is typ-

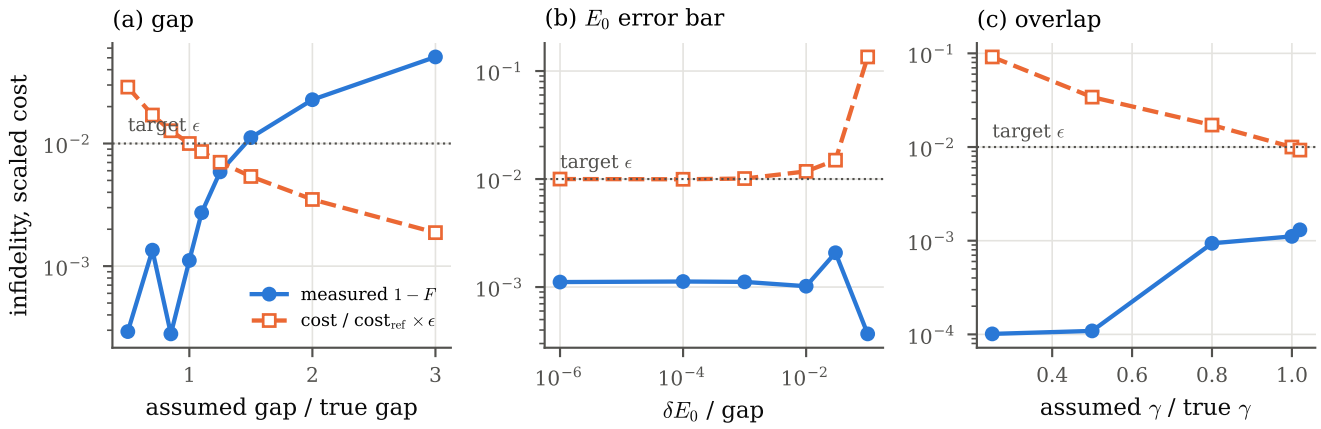


FIG. 5. Sensitivity to the inputs ($N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$). Circles: measured infidelity of the planned circuit on the true trial state. Squares: expected CNOT cost relative to the plan with exact inputs, multiplied by ϵ to share the axis. (a) Assumed gap $f\times$ the true gap; (b) error bar on E_0; (c) assumed overlap $f\times$ the true γ. Source: `results/sensitivity.json`.

ical we repeated the gap and overlap perturbations for every $N \le 14$, all three couplings and the dyadic, optimized and QETU schedules (second order, $\epsilon = 10^{-2}$; Table V). With exact inputs all 36 plans meet the target. An overestimate of the gap by 10% already breaks it in 2 of 36 plans (marginally, worst 1.1×10^{-2}), by 25% in 10 and by 50% in 16. The closed-form dyadic schedule is the most tolerant (0 misses at 25%), because its bound is the loosest; the optimized and Chebyshev filters, whose leakage terms are nearly tight, have no slack to absorb a wrong gap. Overestimating γ broke the target in 12 of 72 plans, all of them through the same mechanism: the inflated γ crossed $1-\epsilon$, so the pipeline dropped the filter altogether (12 of 12). A margin on both inputs is therefore not optional: δE_1 should be a sizeable fraction of the gap when E_1 is not certified, and the empty filter should be accepted only on a certified (for example Eckart) lower bound on γ.

TABLE V. Gap and overlap overestimation for every $N \in \{8, 10, 12, 14\}$ and J_2 (12 instances), $\epsilon = 10^{-2}$, second order: number of plans that miss the target and the worst measured infidelity. f: assumed over true gap or γ. Source: `results/ext_sensitivity.json`.

		dyadic		optimized		QETU	
input	f	miss	worst	miss	worst	miss	worst
gap	1	0/12	1.7×10^{-3}	0/12	4.2×10^{-3}	0/12	2.3×10^{-3}
gap	1.1	0/12	6.5×10^{-3}	1/12	1.1×10^{-2}	1/12	1.0×10^{-2}
gap	1.25	0/12	6.8×10^{-3}	5/12	2.6×10^{-2}	5/12	3.0×10^{-2}
gap	1.5	2/12	6.3×10^{-2}	7/12	6.8×10^{-2}	7/12	9.2×10^{-2}
gap	2	8/12	1.4×10^{-1}	7/12	1.5×10^{-1}	11/12	2.2×10^{-1}
gap	3	11/12	2.2×10^{-1}	12/12	2.5×10^{-1}	12/12	3.1×10^{-1}
γ	1.02	1/12	2.4×10^{-2}	1/12	2.4×10^{-2}	1/12	2.4×10^{-2}
γ	1.05	3/12	5.5×10^{-2}	3/12	5.5×10^{-2}	3/12	5.5×10^{-2}

E. Choosing the ansatz

The trial bond dimension trades circuit depth for overlap. Figure 6 and Table VI show the expected total cost for $\chi = 1$ to 8 (the pipeline choosing the cheaper of the dyadic and optimized schedules). A product state ($\chi = 1$) has an exponentially small overlap and is by far the most expensive start (10^7–10^{23} CNOTs). Going to $\chi = 2$ and $\chi = 4$ lowers the cost by orders of magnitude: at $N = 48$, $J_2 = 0.2411$ the $\chi = 4$ plan needs 9.7×10^6 CNOTs and 2.1×10^8 T gates per success. At $\chi = 8$ the trial state alone satisfies $1 - \gamma \le 10^{-2}$ in 9 of 9 cases ($\gamma = 0.9946$ at $N = 48$, $J_2 = 0.2411$), and the pipeline returns the empty filter: 4,500 CNOTs. For this gapless-to-weakly-gapped one-dimensional chain at $\epsilon = 10^{-2}$, directly compiling a better MPS is therefore much cheaper than filtering a worse one. The filter becomes the better tool when the target is tighter than what a compilable MPS reaches, when a certificate on the output is required, or when the classical method cannot provide a compact high-overlap trial circuit, as for two-dimensional lattices.

F. What sets the cost: gap and overlap

The costs above mix three effects that change together with N: the gap shrinks, the energy scale W grows, and the trial overlap decays. To separate them we hold one input fixed and vary the other (3328 plans, QETU and dyadic, second order, $\epsilon \in \{10^{-2}, 10^{-3}\}$, $N = 8$–48, filter cost only so that the trial circuit does not enter). Both knobs are legitimate inputs to the pipeline: a gap smaller

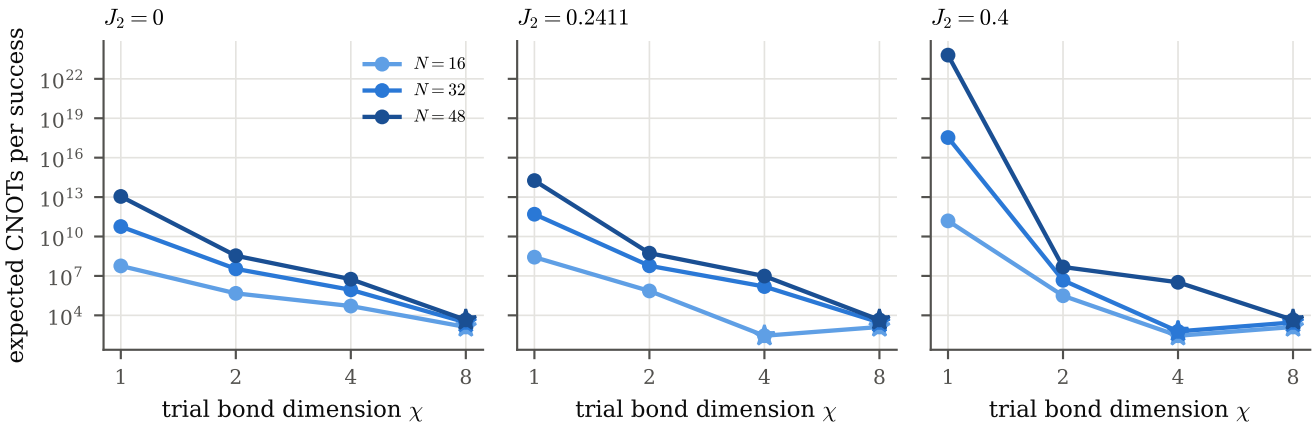


FIG. 6. Expected CNOTs per success versus trial bond dimension ($\epsilon = 10^{-2}$, second order, cheaper of dyadic and optimized). Stars: the trial state alone satisfies $1 - \gamma \leq \epsilon$ and no filter is used. Source: `results/ansatz.json`.

TABLE VI. Overlap γ and expected CNOTs per success for trial bond dimension $\chi = 1, 2, 4, 8$ ($\epsilon = 10^{-2}$, second order, cheaper schedule; no filter when $1 - \gamma \leq \epsilon$).

N	J_2	γ_1	CNOT	γ_2	CNOT	γ_4	CNOT	γ_8	CNOT
16	0	0.028	5.8×10^{7}	0.475	4.7×10^{5}	0.984	5.1×10^{4}	1.000	1.3×10^{3}
16	0.2411	0.019	2.7×10^{8}	0.703	7.2×10^{5}	0.998	2.8×10^{2}	1.000	1.3×10^{3}
16	0.4	0.000	1.5×10^{11}	0.911	3.1×10^{5}	0.999	2.8×10^{2}	1.000	1.3×10^{3}
32	0	0.001	5.7×10^{10}	0.234	3.4×10^{7}	0.913	8.7×10^{5}	0.998	2.9×10^{3}
32	0.2411	0.001	5.0×10^{11}	0.373	5.9×10^{7}	0.972	1.6×10^{6}	0.999	2.9×10^{3}
32	0.4	0.000	3.4×10^{17}	0.768	4.7×10^{6}	0.990	6.0×10^{2}	1.000	2.9×10^{3}
48	0	0.000	1.1×10^{13}	0.125	3.4×10^{8}	0.817	5.6×10^{6}	0.990	4.5×10^{3}
48	0.2411	0.000	1.8×10^{14}	0.191	5.4×10^{8}	0.920	9.7×10^{6}	0.995	4.5×10^{3}
48	0.4	0.000	6.4×10^{23}	0.640	4.8×10^{7}	0.976	3.3×10^{6}	0.998	4.5×10^{3}

than the true one and an overlap smaller than the true one keep the certificate valid.

Overlap at fixed gap. For each N we give the pipeline an overlap γ from 0.01 to 0.97, either with the true gap of that chain or with one common gap for all N (the smallest true gap of the set, at $N = 48$). Between $\gamma = 0.01$ and 0.97 the cost changes by a factor of about 2×10^3 at every N (Fig. 7a). The dependence follows from the design rules: $\eta^2 \propto \gamma\epsilon/(1-\gamma)$ sets $T \propto \Delta^{-1} \ln[(1-\gamma)/(\gamma\epsilon)]$; the second-order step count grows as $T^{3/2}\epsilon_T^{-1/2}$ with $\epsilon_T \propto \sqrt{\gamma\epsilon}$; and the repetitions add $1/p_g \approx 1/\gamma$. Together,

$$\text{cost} \;\propto\; \Delta^{-3/2}\,\gamma^{-5/4}\,\ln^{3/2}\frac{1-\gamma}{\gamma\epsilon}\,\epsilon^{-1/4}. \qquad (11)$$

Fitting the planned costs gives $\gamma^{-1.13}$ (range 1.03–1.19) once the logarithm is divided out, close to the predicted 5/4; as a plain power law over this range the cost goes as $\gamma^{-1.52}$. With the gap held at one common value, the remaining N dependence is $N^{1.6}$ (range 1.4–1.8): the scaled gap $\Delta = g/W$ falls as $1/N$ because $W \propto N$, which alone gives $N^{3/2}$, and the commutator constant and the number of terms per step add a little.

Gap at fixed overlap. We vary the gap in two ways at $\gamma = 0.1$ and 0.5: physically, by scanning $J_2 \in \{0, 0.1, 0.2, 0.2411, 0.3, 0.4, 0.5\}$ at each N (new DMRG and, for $N \leq 14$, ED inputs; ED confirms the triplet as first excitation at every coupling), and at fixed H by handing the pipeline 25–100% of the true gap. The physical scan spans less than one might expect: for $J_2 \leq 0.3$ the finite-size gap barely depends on J_2, and only the dimerized couplings open it at large N (from 0.074 at $J_2 = 0.2411$ to 0.249 at $J_2 = 0.5$ for $N = 48$). All points collapse onto one line in the scaled gap Δ for a given N (Fig. 7b), with cost $\propto \Delta^{-1.50}$ (range 1.44–1.60), the 3/2 of Eq. (11). The gap enters with a larger power than the overlap, but the overlap varies over a much wider range in practice: going from $\chi = 2$ to $\chi = 8$ at $N = 48$ moves γ by a factor of up to 8 (1.6–8), while all couplings together move the gap by a factor of 3.4. To check that the knobs keep the certificate valid we simulated 120 of these plans ($N \leq 14$; $\gamma = 0.03$, 0.3, 0.9 at the true gap, and

the true gap and half of it at $\gamma = 0.5$): all 120 reach the target, with measured infidelity a median 0.06 (at most 0.43) of the certified bound.

G. What the filter actually needs

Section III F describes the certified plan, which assumes the worst case for every input it does not know. To see how the gap and the overlap act on the filter itself, we replace the certificate by simulation for $N = 8$–16 (675 cases). Each trial state is decomposed on the exact eigenbasis (dense diagonalization per S^z block), so the smallest QETU degree $d_{\min}$ that reaches ϵ for that state is found exactly; at that degree we then find the smallest number of Strang steps per query with the Trotterized circuit simulated on the full state vector, allowing a slightly higher degree when it makes one step per query enough. Three families separate the inputs. In the first, $J_2 = 0.2411$ for every N and the gap closes with N. In the second, J_2 is chosen at each N (by bisection on ED) so that the gap is $E_1 - E_0 = 0.40$ exactly ($J_2 = 0.384, 0.510, 0.539, 0.550, 0.555$ for $N = 8$–16). In the third, J_2 is tuned at $N = 10$, 12 and 14 to gaps from 0.30 to 0.55. For the trial states we use the optimal MPS at $\chi = 1$, 2, 4 and synthetic states $\sqrt{\gamma}|E_0\rangle + \sqrt{1-\gamma}|\xi\rangle$ with γ from 0.03 to 0.9, ξ being the normalized excited part of the $\chi = 2$ MPS, so that the overlap changes while the shape of the excited weight does not. $\epsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$.

The degree is set by the gap and the target alone. The measured degrees collapse onto

$$d_{\min}\,\Delta \;\simeq\; \frac{2}{\pi}\cosh^{-1}\frac{1}{\eta}, \qquad \eta^2 = \frac{\gamma\epsilon}{1-\gamma}, \tag{12}$$

for every family, trial and target (Fig. 8b). This is the certified degree with the true gap: the certificate asks for a median 1.09 times $d_{\min}$. Where the excited weight of the trial sits does not help. The Chebyshev filter is equiripple, so every state above the gap is suppressed by the same factor $1/T_d(1/x_s)$, and a trial whose excited weight lies far above the gap needs the same degree as one whose weight sits at the gap. Over the gap scan $d_{\min} \propto \Delta^{-1.00}$.

The certificate's Trotter count is pessimistic, not its filter. One Strang step per query was enough in all 675 cases, against a certified median of 16 steps per query. Commutator bounds are worst-case over all states, but the filter only has to act well on a state supported near the bottom of the spectrum. As a result the certified cost is a median 23 times what the filter needs (range 7–2×10^2), and the real cost follows the degree:

$$\text{cost}_{\text{real}} \;\approx\; \frac{c_{\text{step}}(N)}{\gamma}\,d_{\min} \;\propto\; \frac{c_{\text{step}}(N)\,W}{g\,\gamma}\,\ln\frac{1-\gamma}{\gamma\epsilon}, \tag{13}$$

with c_{step} the CNOTs of one step ($\propto N$) and $g = E_1 - E_0$. Fitting the simulations gives overlap $\gamma^{-0.97}$ once the logarithm is divided out ($\gamma^{-1.21}$ as a plain power, against $\gamma^{-1.45}$ for the certified plan) and gap $\Delta^{-0.98}$ (certified $\Delta^{-1.38}$). With the gap held at 0.40 the cost grows as $N^{2.3}$; at fixed J_2, where g also closes, as $N^{3.1}$ (Fig. 8c). The difference is the $1/g$ of Eq. (13).

Gap and overlap are not independent inputs. Tuning J_2 to hold the gap moves the chain toward the dimerized point $J_2 = 1/2$, where a $\chi = 2$ MPS is close to exact: at the gap-matched couplings the $\chi = 2$ overlap is 0.97–1.00 at every N, against 0.70–0.90 at $J_2 = 0.2411$. Up to $J_2 \approx 1/2$ a larger gap and a better trial come together, so for real trials the two effects compound; beyond it the gap keeps growing while the $\chi = 2$ overlap falls again (to 0.78 at $J_2 = 0.68$, $N = 14$). Holding one of them fixed through J_2 therefore always moves the other, which is why the scans above use synthetic trials.

IV. DISCUSSION

What is guaranteed. Theorem 1 holds whenever its inputs are right. Of these, the spectral upper bound (Weyl) and the filter certificate are rigorous, and the Trotter constants are exact up to the Lanczos tolerance of their largest local norms. The interval for E_0, the lower bound on E_1 and the overlap γ come from DMRG and are well-converged estimates, not bounds; for $N \le 14$ ED confirms them, beyond that the certificate is conditional on all three: the E_0 interval, the gap and the overlap. The gap is the critical input: an overestimate of 10–25% already broke the target for the optimized and Chebyshev filters (Table V), and an overestimated overlap did so whenever it led the pipeline to skip the filter. Rigorous lower bounds on E_1 from classical data (e.g. Temple-type bounds from the energy variance) would remove the gap condition and plug into the same interface. Given such energy bounds, the Eckart inequality $1 - \gamma \le (\langle H\rangle_\psi - E_0)/(E_1 - E_0)$ turns the trial state's energy into a lower bound on γ, which would remove the overlap condition as well.

Where the cost goes. The certified cost is dominated by the Trotter term, whose bound is a median 185× the measured Trotter distance; at the level of step counts the certified second-order plan uses 8.7–39× (median 21) more steps than needed. Second-order formulas recover most of the first-order looseness; bounds restricted to the low-energy subspace [19] or higher-order formulas are the obvious next lever. The leakage term is nearly tight for optimized filters (median 1.4×); for the closed-form schedules it could be tightened by certifying η numerically when L is small enough.

Is the filter worth it here? Not for this model at $\epsilon = 10^{-2}$: a $\chi = 8$ MPS circuit meets the target directly with between one and more than three orders of magnitude fewer CNOTs than the cheapest filtered plan (Table VI). The filter's advantages are a certificate on the output and a cost that depends on the trial overlap only through $\ln(1/\eta)$ and $1/p_g$. The pipeline is generic in H, so the

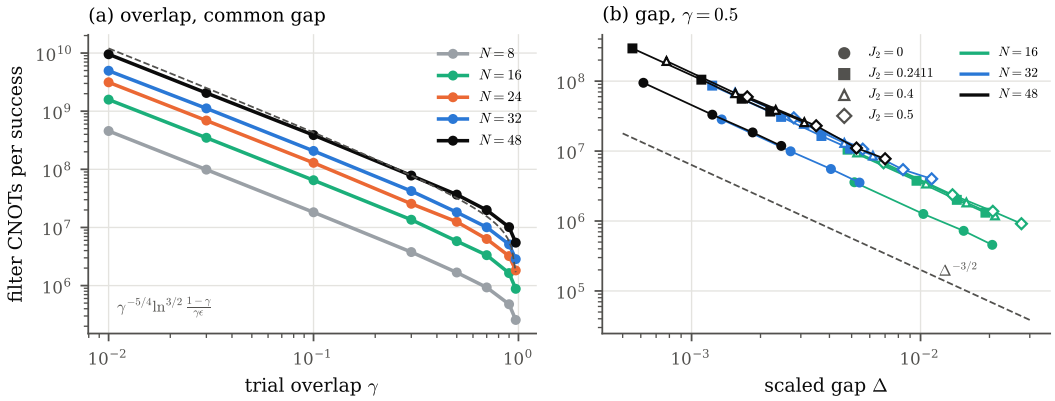


FIG. 7. What sets the filter cost (QETU, second order, $\epsilon = 10^{-2}$, filter only). (a) Expected CNOTs per success versus the trial overlap at one common gap for all N ($J_2 = 0.2411$); dashed: the model of Eq. (11) through the $N = 48$ curve. (b) Versus the scaled gap at $\gamma = 0.5$; each N and J_2 contributes the true gap and 25, 50 and 75% of it. Dashed: $\Delta^{-3/2}$. Source: `results/ext_drivers.json`.

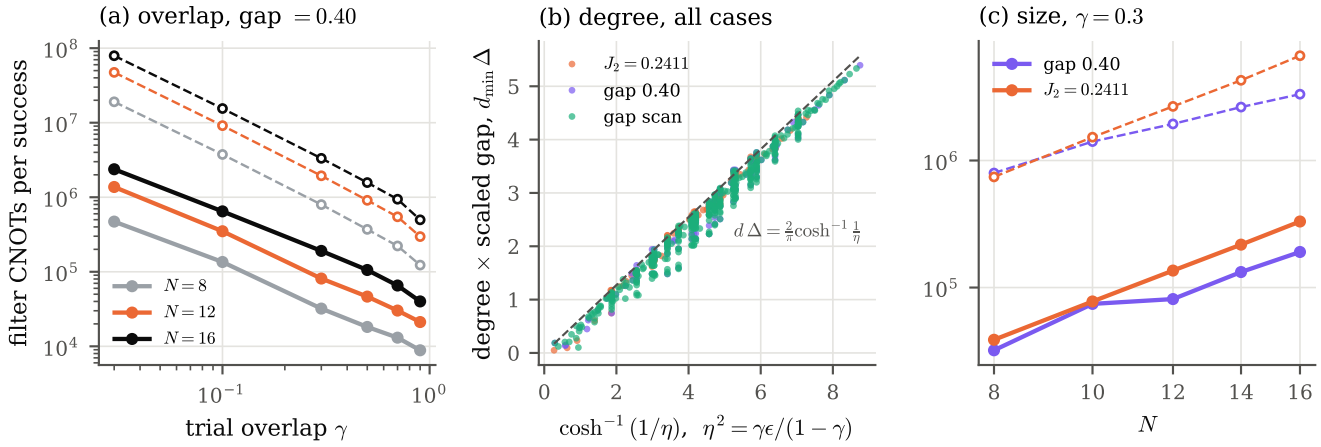


FIG. 8. What the filter actually needs, from exact spectra and state-vector simulation ($N = 8$–16, QETU, second order). (a) CNOTs per success versus overlap at a gap held at 0.40, $\epsilon = 10^{-3}$: simulated (filled) and certified (open, dashed). (b) Smallest degree times scaled gap for all 675 cases against $\cosh^{-1}(1/\eta)$; dashed: Eq. (12). (c) Cost versus N at $\gamma = 0.3$, $\epsilon = 10^{-3}$, with the gap held at 0.40 or J_2 held at 0.2411. Source: `results/emp_trotter.json`, `sbpipe_study/run_empirical.py`.

same comparison can be run for two-dimensional models where MPS compilation scales worse; we leave that to future work.

Limitations. Noiseless simulation; repeat-until-success rather than early abort; ladder CNOT counts (an upper bound, about 1.5× a compiler's); all-to-all connectivity; trial circuits costed by a generic decomposition bound rather than compiled; validation only up to $N = 14$; the optimized designer can fail for small overlaps; E_1 is taken as the lowest $S^z = 1$ state, which is the first excitation in every ED case here but is an assumption for larger N.

CODE AND DATA AVAILABILITY

The `sbpipe` package, the study scripts, all raw results (`sbpipe_study/results/*.json`) and the sources of this paper are available at https://github.com/aaronseymour7/FilterPipeline.

Appendix A: Proof of Theorem 1

Write the trial state as $|\psi\rangle = \sqrt{\gamma'}|g\rangle + \sqrt{1-\gamma'}|\chi\rangle$ with $|g\rangle$ in the ground space (energy $e_0 \in [0, w_0]$ in scaled units), $|\chi\rangle$ orthogonal to it and $\gamma' \geq \gamma$.

Lemma 1 (leakage)**.** *For the exact filter, $v = F(H_s)|\psi\rangle$ has weight outside the ground space at most ℓ, and $\|v\|^2 \geq$*

p_g.

Proof. $F(H_s)$ is a function of H_s, so $v = \sqrt{\gamma'}F(e_0)|g\rangle + \sqrt{1-\gamma'}F(H_s)|\chi\rangle$ with the two parts orthogonal. Every component of $|\chi\rangle$ has energy in $[\Delta, 1]$, so $\|F(H_s)\chi\| \le \sup_{[\Delta,1]}|F| \le \eta F_{\text{pass}} \le \eta|F(e_0)|$. The excited weight is $x/(a+x)$ with $a = \gamma' F(e_0)^2$, $x = (1-\gamma')\|F\chi\|^2 \le (1-\gamma')\eta^2 F(e_0)^2$; it is increasing in x and, after dividing by $F(e_0)^2$, decreasing in γ', which gives ℓ. Finally $\|v\|^2 \ge a \ge \gamma F_{\text{pass}}^2$. □

For pure states the trace distance is $\sin\angle$, so $D(\hat{v}, |g\rangle) = \sqrt{\ell'} \le \sqrt{\ell}$ ($\hat{v} = v/\|v\|$).

Lemma 2 (implementation error)**.** *Let* $B_i = \frac{1}{2}(e^{-i\phi_i}\tilde{S}_i(t_i) + e^{i\phi_i}\tilde{S}_i(-t_i))$ *be the ancilla-0 block of the implemented pulse and* $C_i = \cos(H_s t_i + \phi_i)$*. Then* $\|\prod_i B_i - \prod_i C_i\| \le \epsilon_U$.

Proof. $C_i = \frac{1}{2}(e^{-i\phi_i}e^{-it_iH_s} + e^{i\phi_i}e^{it_iH_s})$, so $\|B_i - C_i\| \le \frac{1}{2}(\|\tilde{S}_i(t_i) - e^{-it_iH_s}\| + \|\tilde{S}_i(-t_i) - e^{it_iH_s}\|) \le \epsilon_i$, where ϵ_i is the product-formula bound (symmetric in the sign of t) plus the synthesis error of the pulse's rotations. Since $\|B_i\|, \|C_i\| \le 1$, telescoping gives $\|\prod B_i - \prod C_i\| \le \sum_i \epsilon_i$. A synthesized trial circuit prepares $|\tilde{\psi}\rangle$ with $\|\tilde{\psi} - \psi\| \le R_{\text{trial}}\delta$, and $\|\prod B_i\| \le 1$, so the total error of the unnormalized output is at most ϵ_U. □

With $\tilde{v} = \prod B_i|\tilde{\psi}\rangle$, $\|\tilde{v} - v\| \le \epsilon_U$. The distance from v to the line through $\tilde{v}$ is at most $\|v - \tilde{v}\|$, so $\sin\angle(v, \tilde{v}) \le \epsilon_U/\|v\| \le \epsilon_U/\sqrt{p_g}$. The triangle inequality for the trace distance gives the bound on D. The states $\hat{v}$, $|g\rangle$ and the normalized output are pure, so $D^2 = 1 - |\langle g|\tilde{\psi}_{\text{out}}\rangle|^2$; since $|g\rangle$ lies in the ground space, $\text{Tr}(\Pi_0\tilde{\rho}) \ge |\langle g|\tilde{\psi}_{\text{out}}\rangle|^2$, which gives (9). (For a mixed output, e.g. under noise, a trace-distance bound alone would not bound the ground-space weight by D^2; this case is outside the theorem.) The success probability is $\|\tilde{v}\|^2 \ge (\|v\| - \epsilon_U)^2 \ge (\sqrt{p_g} - \epsilon_U)^2$. For the energy, the normalized state has weight at least $1 - D^2$ on the ground space and at most D^2 on levels $\le E_{\text{top}}$, so $\langle H\rangle - E_0 \le (E_{\text{top}} - E_0)D^2$. □

Appendix B: Reproducing the results

The code, raw results and paper sources are at https://github.com/aaronseymour7/FilterPipeline. After `pip install -r requirements.txt`, all commands below run from the repository root. `python sbpipe_study/run_study.py all` runs the DMRG front end (cached), the validation (A), scaling (B), sensitivity (C), ansatz (D) and oracle (E) studies and writes `results/*.json`; `python sbpipe_study/make_figures.py` writes every figure, table and the numerical macros used in this text. `python -m pytest sbpipe/tests` runs the component checks (Pauli algebra, Weyl bound, Trotter bounds against dense matrix exponentials, the Dirichlet identity (3), filter certificates against a dense grid, gate counts against an explicit merged sequence and against qiskit, and an end-to-end certificate check).

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